

Boss Challenge — Paper 29 (Class 7)

Soil | The Triangle & its Properties | Respiration in Organisms | Arithmetic Expressions
One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. Rocks broken down very slowly over a long time by the sun, water and wind to form soil is a process called:
(A) sedimentation
(B) weathering
(C) evaporation
(D) erosion
2. The sum of the three interior angles of any triangle always adds up to:
(A) 360°
(B) 270°
(C) 90°
(D) 180°
3. The process by which living cells break down food to release the energy stored in it is called:
(A) respiration
(B) digestion
(C) photosynthesis
(D) excretion
4. In the expression $7 + 3 \times 4$, following the correct order of operations, the part we work out first is:
(A) 7×4
(B) 4 alone
(C) 3×4
(D) $7 + 3$
5. A vertical slice cut down through the ground, showing different layers one below another, is known as the soil:
(A) profile
(B) texture
(C) horizon
(D) crust
6. A triangle whose three sides are all equal in length is known as:
(A) isosceles
(B) scalene
(C) right-angled
(D) equilateral

7. Respiration that uses oxygen to break down glucose completely is called:
- (A) photosynthesis
 - (B) anaerobic respiration
 - (C) aerobic respiration
 - (D) transpiration
8. The value of $7 + 3 \times 4$ is:
- (A) 19
 - (B) 84
 - (C) 40
 - (D) 14
9. Each separate layer that can be seen in a soil profile is called a soil:
- (A) texture
 - (B) grain
 - (C) horizon
 - (D) profile
10. A triangle with exactly two sides equal in length is called:
- (A) isosceles
 - (B) scalene
 - (C) obtuse
 - (D) equilateral
11. Respiration that takes place without using any oxygen is called:
- (A) digestion
 - (B) anaerobic respiration
 - (C) aerobic respiration
 - (D) breathing
12. The value of $(7 + 3) \times 4$, where the bracket is done first, is:
- (A) 14
 - (B) 40
 - (C) 84
 - (D) 19
13. The topmost, dark layer of soil, rich in humus and home to many small living things, is the:
- (A) bedrock
 - (B) subsoil
 - (C) topsoil
 - (D) gravel layer
14. A triangle whose three sides all have different lengths is called:
- (A) scalene
 - (B) equilateral
 - (C) isosceles
 - (D) right

15. During very hard exercise our muscle cells run short of oxygen and partly break glucose down into:
- (A) carbon dioxide and water only
 - (B) glucose
 - (C) lactic acid
 - (D) alcohol
16. In the order of operations, brackets are always worked out:
- (A) last
 - (B) after addition
 - (C) never
 - (D) first
17. The rotted remains of dead plants and animals that make the topsoil dark and fertile are called:
- (A) clay
 - (B) humus
 - (C) gravel
 - (D) bedrock
18. In an equilateral triangle, each of the three interior angles measures exactly:
- (A) 90°
 - (B) 30°
 - (C) 45°
 - (D) 60°
19. The build-up of lactic acid in the muscles after running fast is felt by us as:
- (A) cooling
 - (B) extra energy
 - (C) hunger
 - (D) muscle cramps
20. The value of $20 - 6 \div 2$ is:
- (A) 14
 - (B) 17
 - (C) 10
 - (D) 7
21. The soil that has the largest particles, feels gritty, and lets water drain away the fastest is:
- (A) sandy soil
 - (B) loamy soil
 - (C) humus
 - (D) clayey soil

22. Two angles of a triangle measure 50° and 60° . The third angle measures:

- (A) 70°
- (B) 110°
- (C) 130°
- (D) 80°

23. Yeast carries out anaerobic respiration, breaking sugar down into alcohol and:

- (A) oxygen
- (B) water only
- (C) lactic acid
- (D) carbon dioxide

24. The parts of an expression that are separated from one another by + and - signs are called its:

- (A) factors
- (B) terms
- (C) brackets
- (D) products

25. The soil made of the smallest, tightly packed particles, which holds the most water and feels smooth and sticky when wet, is:

- (A) clayey soil
- (B) loamy soil
- (C) sandy soil
- (D) gravelly soil

26. A triangle that has one of its angles equal to exactly 90° is called a:

- (A) right-angled triangle
- (B) obtuse triangle
- (C) acute triangle
- (D) equilateral triangle

27. Dough rises and bread becomes spongy because the yeast in it releases the gas:

- (A) carbon dioxide
- (B) nitrogen
- (C) hydrogen
- (D) oxygen

28. In the expression $5 + 8 - 2$, the terms are:

- (A) just 11
- (B) 5 and 8 only
- (C) 5×8
- (D) 5, 8 and -2

29. The best soil for growing most crops, a balanced mixture of sand, clay and humus, is called:
- (A) rocky soil
 - (B) clayey soil
 - (C) sandy soil
 - (D) loamy soil
30. In a right-angled triangle, the longest side, which lies opposite the right angle, is called the:
- (A) base
 - (B) altitude
 - (C) hypotenuse
 - (D) median
31. The taking in of air and giving it out, done by the body to exchange gases, is called:
- (A) digestion
 - (B) breathing
 - (C) respiration
 - (D) circulation
32. Without calculating, which is greater: $24 + 17$ or $24 + 15$?
- (A) cannot tell
 - (B) $24 + 15$
 - (C) they are equal
 - (D) $24 + 17$
33. The amount of water a soil can soak up and keep for plants to use is called its water:
- (A) profile
 - (B) percolation rate
 - (C) holding capacity
 - (D) temperature
34. In a right triangle the two shorter sides are 3 cm and 4 cm. The length of the hypotenuse is:
- (A) 12 cm
 - (B) 25 cm
 - (C) 7 cm
 - (D) 5 cm
35. The breathing in of air, when it is drawn into the lungs, is called:
- (A) transpiration
 - (B) exhalation
 - (C) inhalation
 - (D) respiration
36. Without working it out, the value of $38 - 12$ compared with $38 - 20$ is:
- (A) equal to
 - (B) less than
 - (C) greater than
 - (D) cannot be compared

37. How quickly water sinks down and passes on through a soil sample is described by the soil's:
- (A) water-holding capacity
 - (B) humus content
 - (C) colour
 - (D) percolation rate
38. A right triangle has two sides of 6 cm and 8 cm forming the right angle. Its hypotenuse is:
- (A) 100 cm
 - (B) 10 cm
 - (C) 48 cm
 - (D) 14 cm
39. The breathing out of air, when it is pushed out of the lungs, is called:
- (A) inhalation
 - (B) exhalation
 - (C) percolation
 - (D) digestion
40. Using the distributive idea, 6×13 can be rewritten as:
- (A) $6 + 10 + 3$
 - (B) $6 \times 10 \times 3$
 - (C) $6 \times 10 + 6 \times 3$
 - (D) $6 \times 10 - 3$
41. 200 mL of water takes 40 minutes to percolate through a soil sample. Its percolation rate, in mL per minute, is:
- (A) 160
 - (B) 8
 - (C) 240
 - (D) 5
42. An exterior angle of a triangle is equal to the sum of the two:
- (A) all three interior angles
 - (B) base angles
 - (C) adjacent angles
 - (D) opposite interior angles
43. Compared with the air we breathe in, the air we breathe out has much more of the gas:
- (A) nitrogen
 - (B) oxygen
 - (C) carbon dioxide
 - (D) hydrogen

44. By the distributive law, 8×99 is easiest to work out as:

- (A) $8 \times 100 + 8$
- (B) $8 \times 100 - 8$
- (C) $8 + 99$
- (D) $8 \times 90 + 9$

45. One soil lets 300 mL of water through in 30 minutes; another lets 300 mL through in 60 minutes. The faster-draining soil is the one with the:

- (A) shorter percolation time
- (B) more humus
- (C) darker colour
- (D) larger water-holding capacity

46. The two interior opposite angles of a triangle are 45° and 55° . The exterior angle at the third vertex is:

- (A) 145°
- (B) 80°
- (C) 100°
- (D) 90°

47. The large, dome-shaped muscle lying below the lungs that helps in breathing is the:

- (A) heart
- (B) diaphragm
- (C) ribs
- (D) liver

48. The value of 8×99 is:

- (A) 792
- (B) 808
- (C) 891
- (D) 729

49. Heavy water-holding soils such as clayey soil and clay-rich loam are best for growing crops like:

- (A) nothing at all
- (B) cactus
- (C) only mushrooms
- (D) wheat and gram

50. A segment drawn from a vertex of a triangle straight to the midpoint of the opposite side is a:

- (A) median
- (B) altitude
- (C) angle bisector
- (D) hypotenuse

51. When we breathe in, the diaphragm moves down and flattens, while the ribs move:
- (A) in only
 - (B) down and in
 - (C) up and out
 - (D) not at all
52. Changing the order of the two numbers, $15 + 27$ gives exactly the same answer as:
- (A) 27×15
 - (B) $27 - 15$
 - (C) $27 + 15$
 - (D) $15 - 27$
53. Light, well-draining sandy-loam soil is the right kind for growing the crop:
- (A) cotton
 - (B) nothing
 - (C) paddy
 - (D) water lilies
54. The perpendicular line segment drawn from a vertex straight to the opposite side of a triangle is called an:
- (A) exterior angle
 - (B) altitude
 - (C) base
 - (D) median
55. The number of times a person breathes in and out in one minute is called the:
- (A) percolation rate
 - (B) pulse
 - (C) heart rate
 - (D) breathing rate
56. Swapping the two numbers in $20 - 8$ to make $8 - 20$ changes the answer, which shows that subtraction is:
- (A) always zero
 - (B) the same either way
 - (C) not commutative
 - (D) commutative
57. The removal of the fertile top layer of soil by wind or flowing water is called soil:
- (A) erosion
 - (B) weathering
 - (C) pollution
 - (D) percolation

58. The number of medians that a triangle has is:

- (A) 1
- (B) 6
- (C) 2
- (D) 3

59. If a person breathes 18 times each minute, the number of breaths taken in 3 minutes is:

- (A) 15
- (B) 21
- (C) 54
- (D) 6

60. Removing the brackets, the value of $50 - (20 + 5)$ is:

- (A) 30
- (B) 25
- (C) 35
- (D) 75

61. Planting trees and grasses protects land from erosion mainly because their roots:

- (A) bind the soil particles together
- (B) change the soil colour
- (C) add water to the soil
- (D) warm the soil

62. For three given lengths to form a triangle, the sum of any two of the sides must be:

- (A) less than the third side
- (B) greater than the third side
- (C) equal to the third side
- (D) equal to 180°

63. Fish take in the oxygen dissolved in water using their:

- (A) nostrils
- (B) gills
- (C) lungs
- (D) skin

64. The value of $30 - (10 - 4)$ is:

- (A) 20
- (B) 16
- (C) 36
- (D) 24

65. Throwing harmful chemicals, plastic bags and other waste onto the land, which spoils the soil, is called soil:
- (A) weathering
 - (B) pollution
 - (C) erosion
 - (D) percolation
66. Which of the following sets of lengths can form a triangle?
- (A) 1 cm, 1 cm, 5 cm
 - (B) 2 cm, 3 cm, 8 cm
 - (C) 5 cm, 6 cm, 9 cm
 - (D) 4 cm, 4 cm, 9 cm
67. An earthworm has no lungs or gills; it breathes through its moist:
- (A) nostrils
 - (B) leaves
 - (C) gills
 - (D) skin
68. A soil sample lets 50 mL of water pass in each 10-minute period, for 4 such periods. The expression for the total water passed is:
- (A) $50 - 4$
 - (B) 50×4
 - (C) $50 + 4$
 - (D) $50 / 4$
69. Below the topsoil lies a harder, lighter layer with little humus but more minerals, known as the:
- (A) subsoil
 - (B) topsoil
 - (C) humus layer
 - (D) bedrock
70. In an isosceles triangle, the two angles that lie opposite the two equal sides are:
- (A) both 90°
 - (B) different
 - (C) equal to each other
 - (D) always 60°
71. Insects such as cockroaches take air into the body through tiny holes along their sides called:
- (A) nostrils
 - (B) stomata
 - (C) gills
 - (D) spiracles

72. A boy breathes 20 times a minute. The expression for the breaths he takes in 5 minutes plus 3 extra breaths is:

- (A) $20 \times (5 + 3)$
- (B) $20 \times 5 \times 3$
- (C) $20 \times 5 + 3$
- (D) $20 + 5 + 3$

73. The bottommost layer of the soil profile, made of solid, continuous rock, is the:

- (A) topsoil
- (B) bedrock
- (C) humus
- (D) subsoil

74. An isosceles triangle has its two equal angles each measuring 70° . The third angle measures:

- (A) 40°
- (B) 110°
- (C) 70°
- (D) 50°

75. Plants take in and give out gases for respiration mainly through tiny pores on their leaves called:

- (A) spiracles
- (B) roots
- (C) stomata
- (D) gills

76. The value of $20 \times 5 + 3$ is:

- (A) 103
- (B) 28
- (C) 160
- (D) 130

77. The size of the particles that make up a soil decides the soil's:

- (A) erosion
- (B) humus
- (C) profile
- (D) texture

78. An equilateral triangle also counts as a special isosceles triangle because it has:

- (A) only one line of symmetry
- (B) a right angle
- (C) all different sides
- (D) at least two equal sides

79. The wind-pipe that carries air from the nose and throat down towards the lungs is the:
- (A) trachea
 - (B) diaphragm
 - (C) oesophagus
 - (D) aorta
80. Which sign makes the statement true: $12 + 8$ ____ 5×4 ?
- (A) none
 - (B) $>$
 - (C) $=$
 - (D) $<$
81. In heavy rain, sandy soil floods far less than clayey soil because sandy soil has a:
- (A) smaller particles
 - (B) lower percolation rate
 - (C) higher percolation rate
 - (D) more humus
82. The three angles of a triangle are in the ratio 1 : 2 : 3. The largest of these angles is:
- (A) 180°
 - (B) 60°
 - (C) 120°
 - (D) 90°
83. The pair of organs in our chest where oxygen passes from the air into the blood are the:
- (A) lungs
 - (B) gills
 - (C) kidneys
 - (D) liver
84. The value of $3 \times (4 + 2) - 5$ is:
- (A) 13
 - (B) 9
 - (C) 15
 - (D) 21
85. Earthworms are called a farmer's friend because, as they tunnel through the ground, they:
- (A) remove all the humus
 - (B) make burrows that let air and water in
 - (C) harden the soil
 - (D) eat the crop roots
86. Can a single triangle have two right angles?
- (A) no, the angle sum would already exceed 180°
 - (B) only if equilateral
 - (C) yes, if it is large
 - (D) yes, always

87. The roots of a plant get the oxygen they need for respiration from the:
- (A) air spaces between soil particles
 - (B) leaves only
 - (C) sunlight
 - (D) water in the clouds
88. Regrouping the numbers, $(12 + 8) + 5$ gives the same total as:
- (A) $12 + 8 - 5$
 - (B) $(12 \times 8) + 5$
 - (C) $12 + (8 + 5)$
 - (D) $12 - (8 + 5)$
89. The ideal field soil mixes the fast drainage of sand with the water-holding of clay; this balanced soil is:
- (A) gravel
 - (B) loam
 - (C) pure sand
 - (D) pure clay
90. Within a single triangle, you can have at most one:
- (A) vertex
 - (B) acute angle
 - (C) obtuse angle
 - (D) side
91. The word equation for aerobic respiration is: glucose + oxygen $\rightarrow$ carbon dioxide + water +:
- (A) alcohol
 - (B) sunlight
 - (C) lactic acid
 - (D) energy
92. To make 6×47 easier, splitting 47 into $40 + 7$ and multiplying each part by 6 uses the:
- (A) order of brackets only
 - (B) distributive law
 - (C) no rule at all
 - (D) commutative law of subtraction
93. Warming a small lump of moist soil and seeing water droplets form on the cool lid shows the soil contains:
- (A) humus
 - (B) moisture
 - (C) salt only
 - (D) only air

94. In a right-angled triangle, the two sides that form the right angle serve as the base and the:

- (A) hypotenuse
- (B) median
- (C) exterior angle
- (D) height

95. Lime water turns milky when we blow our breath through it, which proves that exhaled air contains:

- (A) hydrogen
- (B) carbon dioxide
- (C) oxygen
- (D) nitrogen

96. In the expression $100 - 25 \times 3$, the very first step is to work out:

- (A) $25 + 3$
- (B) 100×3
- (C) 25×3
- (D) $100 - 25$

97. Pouring water onto a lump of dry soil and seeing bubbles rise out of it shows the soil contains:

- (A) humus
- (B) air
- (C) water
- (D) only minerals

98. In any triangle, the longest side always lies opposite the:

- (A) largest angle
- (B) smallest angle
- (C) shortest side
- (D) right angle only

99. Compared with our breathing while we sleep, our breathing rate while we are running:

- (A) stops
- (B) increases
- (C) decreases
- (D) stays exactly the same

100. The value of $100 - 25 \times 3$ is:

- (A) 225
- (B) 75
- (C) 50
- (D) 25